08.05.2025(E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025 –April 2025		
Maximum Marks: 100	**Examination: ESE Examination**	**Duration:3 Hrs.**
Programme code: 01 **Programme: B. Tech in Computer Engineering**	**Class: SY**	**Semester: IV(SVU 2020)**
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department:** COMP	
Course Code: 116U01C404	**Name of the Course: Theory of Automata with Compiler Design**	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	Define the various phases of compiler.	5
ii)	Examine the Closure properties of Regular Language with respect to Union and Concatenation.	5
iii)	Consider the grammar given below, Eliminate the Unit Production from it. S->AB A->a B->C|b C->D D->E E->a	5
iv)	State the 7 tuple Formal Definition of Turing Machine	5
v)	Solve the Post Correspondence Problem for the two lists given below and give the solution sequence. List1=[1,10111,10] List2=[111,10,0]	5
vi)	State the Rice theorem.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Construct a Deterministic Finite Automata equivalent to Machine M=({qo,q1,q2},{a,b}, δ,q0,{q2}), Transition Table-	5

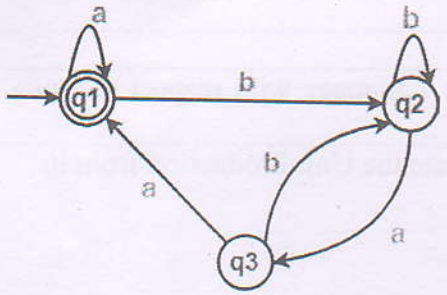
States	a	B
->q0	q0	q0,q1
q1	q2	q2
*q2	-	-

ii) Construct the Moore Machine which is equivalent to the Mealy Machine given below. Show all the intermediate steps. Draw the final transition diagram as well. 5

	Inputs			
	0		1	
States	Q	Δ	Q	Δ
q1	q3	0	q2	0
q2	q1	1	q4	0
q3	q2	1	q1	1
q4	q4	1	q3	0

OR

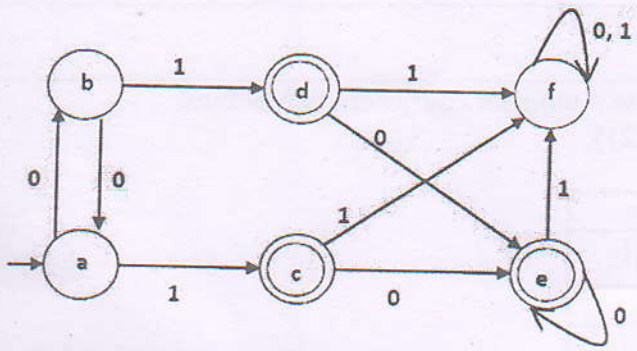
Q2 A Find Regular Expression for the following Automata using Arden's Theorem: 10



Q 2 B Solve any **One** **10**

i) Write the Regular Expressions for the three languages, L1, L2 and L3. 10
L1 represents the set of strings with alphabet set ∑={a,b} such that each word begins and ends with "a".
L2 is the set of all strings such that no of occurrences of "a" is divisible by 3 for ∑={a,b,c}.
L3 is the set of all strings with at the most 2 zeros over ∑={0,1}.

ii) Minimize the DFA given below using Myhill Nerode Theorem i.e. Box Method, with Start State as "a". 10



Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Construct CFGs for the following grammars:-	10

	b) $L(G)=\{ a^m b^n \mid m!=n \}$	
ii)	Convert the grammar with the following productions into CNF: S->aSb|aA|bB|a|b A->aA|a B->bB|b Show all the intermediate steps.	10
iii)	Consider the context free grammar G with the following productions:- S->0S0|1S1|A A->2B3 B->2B3|3 Describe the language generated by the above grammar Further, For the string "0100123310010", Derive the Leftmost Derivation.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Design a PDA for accepting the language $L=\{a^n b^{2n} \mid n>=1\}$. Also draw the transition diagram and Show the simulation using Instantaneous description.	10
ii)	Construct a TM to replace each occurrence of 'a' by 'b' for the strings over $\Sigma=\{a,b\}$. Also draw the transition diagram and Show the Simulation with an example.	10
iii)	Design a TM to accept the language containing substring "aba" over $\Sigma=\{a,b\}$. Also draw the transition diagram and Show the Simulation with an example.	10

Que. No.	Question	Max. Marks
Q5	Solve any **four**	**20**
i)	Design a DFA that reads strings made up of letters in the word 'COMPUTER' and recognizes those strings that contain the word 'PURE' as a substring.	5
ii)	Find the reduced grammar that is equivalent to the CFG given below by removing useless productions:- S->aC|SB A->bSCa B->aSB|bBC C->aBC|ad	5
iii)	Consider the following grammar S->ABAC A->aA|ε B->bB| ε C->c Eliminate all the ε-productions from the grammar without changing the language	5
iv)	Illustrate the working of Multi Tape Turing Machine with the help of suitable diagrams.	5
v)	Examine the closure properties of Recursive Languages	5
vi)	Differentiate between Recursive Enumerable Language and Recursive Language.	5